

Smart Grids: A Must For India

Smart grids would help India march faster along the path of development. However, implementing smart grids in our country is no mean feat. Why? And what steps has the government taken to overcome these challenges?

The total installed capacity of power generation in India is about 160 giga-watts (GW) and the target for 2012 is about 200 GW. Our country is facing huge aggregate technical and commercial (AT&C) losses, the major contributions being technical losses, theft, billing and collection," says Ramesh.

India's transmission and distribution losses are amongst the highest in the world, averaging 26 per cent of total electricity production; in some states, the figure is as high as 62 per cent. When non-technical losses such as energy theft are included in the total, average losses are as high as 50 per cent. The financial loss has been estimated at 1.5 per cent of the national GDP, and is growing steadily. Ramesh opines that smart grids, doubtlessly, are the solution to such problems.

"The smart grid is a new way to use a network to optimise the delivery of energy to consumers to manage efficiency, reduce cost and increase reliability," adds Raina.

Miles to go

However, the mere availability of knowledge and products does not make the implementation of smart grids any easier in a country like India where the power sector is quite a mess. Several impediments lie along the way: inefficiency of the energy system, complex integration of alternative, distributed power sources; lack of common management, visibility and coordinated control; lack of reliability and resilience.

"Another problem in the implementation of smart grids in India is safeguarding our equipment from unsocial elements. Vandalism, sabotage of the system is common. Damaging the meters and other equipment provided on poles and transformers is not uncommon," says Ramesh.

Possible, with commitment

The Indian government has realised this and taken a major step in the form of the Restructured Accelerated Power Development Reforms Programme (RAP-DRP), which aims to first clear up the existing system to make it amenable for smart grids and then implement smart grids. Various tasks under the programme include consumer indexing, GIS and asset mapping, metering of distribution transformers and feeders, automatic data logging, feeder segregation and ring fencing, IT applications for prompt response to consumer grievances, meter reading, billing and collection, energy audits and establishment of a base line data system. Other tasks include renovation, modernisation and strengthening of 11kV sub-stations, transformers and transformer centres, reconductoring of



Picture shot by S. Ramesh at Karol Bagh, New Delhi, portrays the current disorderliness in the power sector

lines at the 11kV level and below, load bifurcation and balancing, strengthening at 33kV and 66kV levels, and installation of capacitor banks, mobile service centres, etc.

One step ahead, some smart grid pilots are under way in states like Maharashtra, Karnataka and New Delhi, in association with companies like Larsen & Toubro, Telvent, GE, Cisco, IBM and Tata Power.

The government has also launched a taskforce and a forum for smart grid development. The forum is a non-profit voluntary consortium of public and private stakeholders, research institutions and selected utilities with the objective of accelerating smart grid development in India, while the taskforce is going to be an inter-ministerial group to ensure awareness, coordination and integration of the diverse activities related to smart grid technologies, promotion of smart grid research and development, collaboration on interoperability framework and so on.

This is an overall sign of the country's commitment towards the cause.

Some of India's smart grid pilots

North Delhi Power Limited (NDPL) has implemented a smart grid project in association with Tata Power. It uses products such as GE's PowerOn coupled with automatic metering infrastructure (AMI), and GE's SmallWorld geographic information system. NDPL seems to have reduced losses from 54 to 18 per cent in the past few years.

Karnataka has planned a pilot at Bengaluru's Electronic City. A budget of one billion rupees has been assigned, and the project plan was drafted in January this year. The implementation is expected to take two years.

Maharashtra has also planned a project with Larsen & Toubro and Telvent.